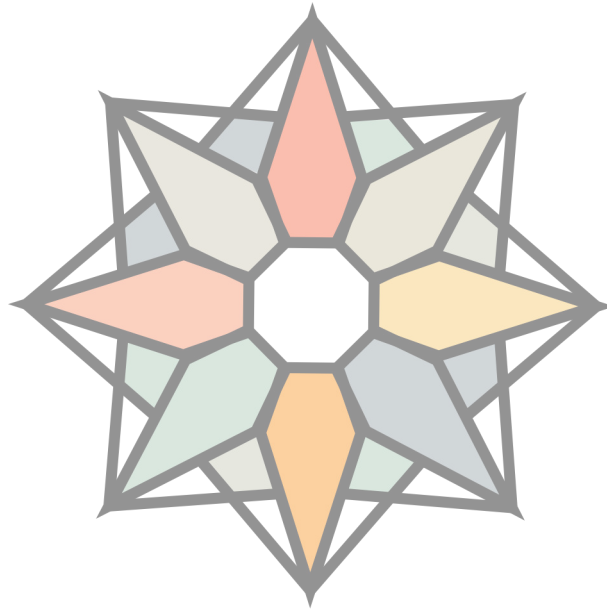




## **Stained Glass Panels**

# **Hackathon Grasshopper Users Meeting**

**Ewa Romanczuk**

**Bartłomiej Tokarzewski**

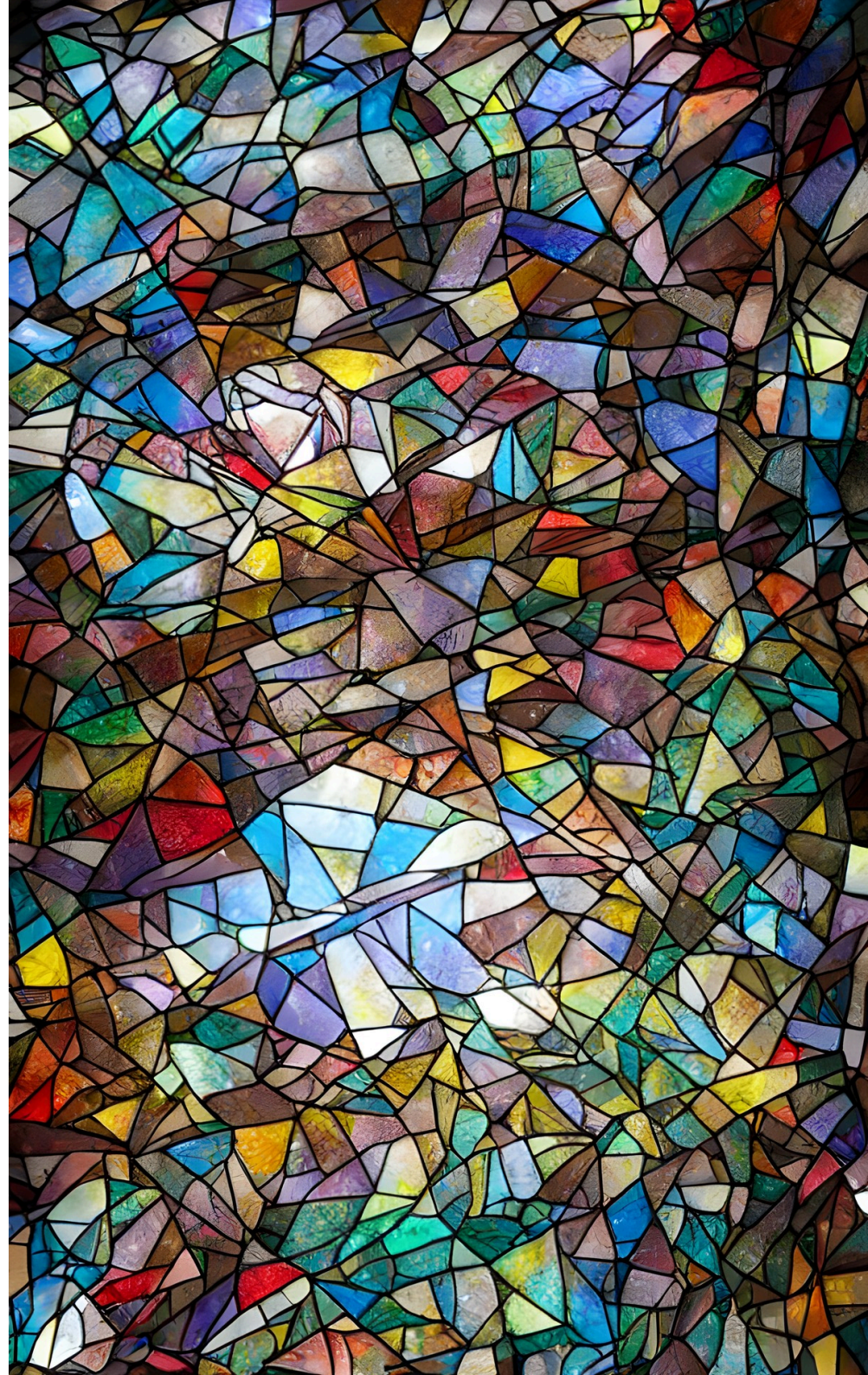
**Martyna Hypta**

**Mateusz Tokarczyk**

**Somia Hassan**

# Content

- The Problem
- The Vision
- The Inputs
- The Workflow
- The Tools
- Live Demo
- What's Next





# The Problem

**We want unique and beautiful elements in our spaces.**

**Problem:** Custom ornament is expensive because it requires intense manual design and specialized craftsmanship.

**Result:** Budget constraints force us back into sterile, mass-produced minimalism.



# Our Solution

**We move the complexity from manual labor to software.**

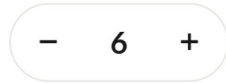
**Platform:** A configurator that lets anyone turn an image into a stained glass panel.

**Output:** Instant, 3D-printable files. No specialized craft skills required; no fortune needed.

# The Inputs



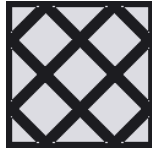
Raster



Number of colours



Colours



Pattern of the panels

# The Outputs

Visualisation

Model 3D in .glb

# The workflow

Parallel tracks from the start. We split the team into:

- image-processing - Somia, Mateusz
- geometry - Martyna, Ewa
- web/render/deploy - Bartłomiej
- UI and UX design - Martyna

# The Tools



**Rhino 7 + Grasshopper**

First geometry engine — image sampling, colour quantization, panelization, BREP frames/extrusion



**Python / Jupyter**

Early model transport from Grasshopper to the web viewer



**Rust + wasm-bindgen**

The production engine, compiled to WebAssembly (sg-wasm/)



**three.js (WebGPU)**

The 3D "beauty" renderer



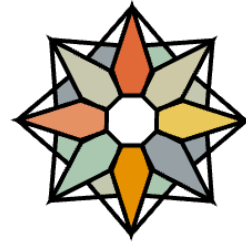
**Illustrator/Photoshop**

Logo (star rosette), grasshopper mascot, loading GIF, app mockups



**Cloudflare + Wrangler**

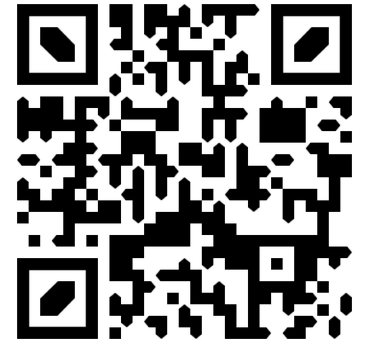
Hosting (no-build static site)



# STAINED GLASS

*Pick your colours. Pick your pattern. Let the light do the rest.*

Stained glass used to be the language of cathedrals and grand estates.  
We're bringing it back — to your home, your studio, your space. Design  
your own piece, choose your colours and pattern, and we'll print it. No  
craft skills needed. No fortune required.

[HOW IT WORKS](#)[MAKE MY ART](#)



# What's Next?

Our prototype can develop in multiple ways, we propose this roadmap to advance our Idea:

- 1) Create viable way of connecting panels to each other using Printable frame exported using website
- 2) Line recognition of pictures uploaded on website
- 3) Optimised technical view with detail of connections and 2d drawings
- 4) The printer selection
- 5) Estimated price of the finished product
- 6) Gallery of the pictures with finished model
- 7) Online shop for special orders and infrastructure
- 8) 3D printer farms for orders
- 9) Metal printing of the mesh

**Thank you!**